# TARUN REDDY

Willing to relocate / open to visa sponsorship | +1 (351) 220-1157 | tarunreddy2811@gmail.com | LinkedIn | GitHub | Portfolio

## SUMMARY

Machine Learning / AI Engineer with hands-on experience building end-to-end AI systems, from real-time computer vision pipelines to LLM-powered SQL agents and audio intelligence platforms. Co-founder of InsightScribe, a production-grade AI platform using OpenAI Whisper and LangChain. Strong foundation in model optimization, NLP, and deploying AI features via REST APIs.

## EDUCATION

**Master of Science in Computer Science — Clark University**
*Aug 2024 – May 2026*
GPA: **3.7 / 4.0**

**Bachelor of Technology in Computer Science — JNTU**
*Aug 2019 – May 2023*
GPA: **3.5 / 4.0**

## PROFESSIONAL EXPERIENCE

### Teaching Assistant — Clark University

*Aug 2025 – Expected May 2026 | Worcester, MA*

- Mentored graduate students in machine learning concepts, Python environments, and full-stack integration
- Assisted with debugging ML pipelines, dependency issues, and model execution workflows
- Explained deep learning and system design concepts clearly to diverse technical audiences

### AI/ML Engineering Intern — IMAI Innovations

*Dec 2023 – May 2024 | Abu Dhabi, UAE (Remote)*

- Optimized model architecture and training workflows, reducing training time by **20%** while maintaining inference accuracy
- Built ML-driven HR automation tools that reduced manual PeopleOps workload by **40%**
- Engineered end-to-end ML pipelines including data preprocessing, feature engineering, model training, and evaluation
- Deployed AI-driven features via **secure REST APIs**, enabling seamless integration with client-facing platforms
- Evaluated model performance using precision, recall, F1-score, and ROC-AUC to ensure reliability
- Analyzed trade-offs between model complexity, latency, and accuracy to support deployment in resource-constrained environments

## PROJECTS

### InsightScribe - AI Intelligence Platform | GitHub

*(Co-Founder & ML/AI Engineer | In Development)*

- Built audio transcription and NLP pipeline using **OpenAI Whisper and HuggingFace**, processing uploaded audio into structured, searchable transcripts
- Engineered cross-recording intelligence system to surface patterns and insights across multiple audio sessions
- Designed and integrated summarization and insight extraction features using **LLM-based pipelines**
- Collaborated in a co-founder engineering environment, owning the full ML/AI layer of a production-grade platform

### Real-Time Facial Emotion & Age Detection System | GitHub

*(Independent ML Project | End-to-End Ownership)*

- Built a real-time full-stack application using **OpenCV, DeepFace, and Python**, integrating live webcam feeds with computer vision models through a clean software pipeline
- Engineered low-latency data processing architecture to handle continuous video stream input, achieving real-time inference with minimal frame drop
- Integrated a custom **CNN-based age prediction model** into the application layer, improving classification accuracy by **18%** through preprocessing optimizations
- Designed modular, maintainable code structure separating video capture, model inference, and UI overlay, demonstrating production-oriented software engineering practices.

### EnterpriseIQ - LangChain SQL AI Agent | GitHub

*(Independent Project)*

- Built a production-style **LLM-powered SQL agent** using **LangChain and Groq's LLaMA-3.3-70b**, enabling natural language querying of enterprise databases without SQL knowledge
- Engineered agent reasoning pipeline using LangChain's `create_sql_agent` with automatic **schema introspection**, handling complex multi-table JOINs and aggregations autonomously
- Designed prompt flow and tool-calling logic to resolve ambiguous business questions into accurate, context-aware SQL queries
- Built interactive **Streamli** interface with suggested query prompts, delivering a non-technical user-facing AI product end-to-end

## SKILLS

**Programming:** Python, SQL, C, R
**ML & AI:** Machine Learning, Deep Learning, TensorFlow, Scikit-learn, NLP, Computer Vision, CNNs, LSTMs, LLMs
**Data & Tools:** Pandas, NumPy, Jupyter Notebook, PyTorch
**Engineering:** REST APIs, Git, Modular Code, Model Evaluation, Model optimization
**Cloud:** AWS, Google Cloud

## CERTIFICATIONS